

- *Searching for a pattern:*

```
[testuser@localhost ~]$ pat="/sbin/nologin"
```

```
[testuser@localhost ~]$ awk -F" " -v pat=${pat} 'NF~pat{print $1}' passwd2
```

This provides a list of all the users who are not allowed to log in to the system via the console, telnet, *ssh*, etc. We specify the field separator as ":" using the *-F* switch and pass the shell variable *pat* as pattern to Gawk using the *-v* switch. We match the pattern against the last field (\$NF) of each record and print the user-name(\$1 is the 1st field) for the records that match the given pattern. While matching *\$NF* against the variable *pat*, we use *pat* and not *\$pat* since the latter will look for a field with the field number as *"\$bin/nologin"* which will be a null value.

- **Inserting text:** Let's suppose, we want to add a comment "Not allowed to log in" to the `passwd2` file for all users that have `"/sbin/nologin"` as shell, and "Allowed to log in" for the rest. We know that comments can be placed in the `GEOS` field in the `passwd` file, which is the fifth field in the file. Then, we can use the following code to add the comment:

```
awk -F: -v pat=$(cat) -v OFS=":" {
    if ($NF==pat) $5="Not allowed to Login";
    else $5="Allowed to login";
    print $0 passwd2
```

Building on the previous example, we set `OFS=":"`. The default value for `OFS` is space and the `passwd` file has `:` as the separator.

- *Replacing text:* In this example, we will try to replace `/bin/bash` with `/bin/sh`.

```
pat="/bin/bash"
awk -F"." -v pat=${pat} -v OFS="." 'BEGIN{newshell="/bin/\sh"}
{sub(pat,newshell,$NF); print}' passwd2
```

We utilise the sub-function to replace `/bin/bash` with `/bin/sh`. We also use the special `BEGIN` pattern to initialise the newshell variable.

- *Range pattern*: Users with UID between 500 and 510.

```
awk -F"." -v pat=$(pat) -v OFS=":" '{ $3~/500/,$3~/510/' passwd2
```

In this first article we covered the basic concepts of Awk. In subsequent articles we will cover advanced topics like arrays, user-defined functions, in-built functions available for string manipulation, and date functions. We will also look at some useful real-world examples like extracting a particular table from a MySQL dump, calculating the date for a few days before or after the current date, and formatting the output.

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